

SHANGHAI BAOSHAN, China

WMO: 583620

Lat: 31.3908N

Lon: 121.4447E

Elev: 9

StdP: 101.22

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.9	(c) -0.6	(d) -12.1	(e) 1.3	(f) 2.0	(g) -9.6	(h) 1.7	(i) 2.6	(j) 7.7	(k) 4.1	(l) 7.1	(m) 4.1	(n) 2.6	(o) 270	(p) 0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.8	(c) 35.5	(d) 26.7	(e) 34.2	(f) 26.5	(g) 33.0	(h) 26.2	(i) 27.8	(j) 32.5	(k) 27.4	(l) 31.8	(m) 26.9	(n) 31.2	(o) 3.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.2	30.3	26.2	21.6	29.9	25.8	21.1	29.6	89.2	32.7	87.0	32.0	85.1	31.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.5	5.9	-4.1	37.2	1.4	1.2	-5.1	38.1	-5.9	38.8	-6.7	39.4	-7.7	40.3
			-5.8	28.7	1.6	0.7	-6.9	29.2	-7.8	29.6	-8.7	30.0	-9.9	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	5.1	6.5	10.5	16.0	21.1	24.5	29.2	28.8	24.9	19.9	14.0	7.7	(6)	(7)
	DBStd	8.84	3.22	3.46	3.63	3.53	2.73	2.67	2.73	2.28	2.47	2.68	3.65	3.53	(7)	(8)
	HDD10.0	400	157	107	37	2	0	0	0	0	0	0	8	88	(8)	(9)
	HDD18.3	1564	411	331	244	88	6	0	0	0	0	15	137	331	(9)	(10)
	CDD10.0	3098	4	9	52	180	343	435	594	583	448	306	127	16	(10)	(11)
	CDD18.3	1221	0	0	1	17	91	185	336	324	198	62	6	0	(11)	(12)
	CDH23.3	11717	0	1	9	106	481	1309	4266	3944	1440	151	9	0	(12)	(13)
	CDH26.7	4725	0	0	1	22	120	405	2102	1713	346	16	0	0	(13)	(14)
(14) Wind	WSAvg	3.0	2.9	3.0	3.1	3.1	3.1	2.9	3.1	3.4	3.1	2.8	2.8	2.9	(14)	(15)
(15) Precipitation	PrecAvg	1148	51	59	84	92	105	164	131	169	138	61	58	42	(15)	(16)
	PrecMax	1672	196	143	173	193	242	486	361	455	352	296	119	155	(16)	(17)
	PrecMin	756	0	8	13	30	22	37	24	0	12	1	2	0	(17)	(18)
	PrecStd	220	35	35	38	41	49	85	79	113	86	57	37	35	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.8	24.4	29.5	32.2	35.0	37.4	36.9	33.9	28.6	24.6	18.4	(19)	(20)
		MCWB	12.8	14.6	15.9	19.9	20.1	25.9	26.5	26.8	26.0	22.6	19.0	14.5	(20)	(21)
	2%	DB	13.5	16.5	21.1	26.3	29.7	32.7	36.0	35.0	31.6	26.3	22.3	16.1	(21)	(22)
		MCWB	10.2	12.2	14.2	18.2	19.9	25.4	26.5	26.7	25.5	20.9	18.0	12.6	(22)	(23)
	5%	DB	11.4	13.7	18.7	23.9	27.7	30.7	34.8	33.7	29.8	24.9	20.7	14.3	(23)	(24)
		MCWB	8.8	10.2	13.0	17.6	19.9	24.6	26.7	26.5	25.0	20.0	16.9	11.4	(24)	(25)
	10%	DB	9.7	11.5	16.4	21.7	26.0	28.9	33.6	32.5	28.5	23.7	19.1	12.8	(25)	(26)
		MCWB	7.4	8.7	12.1	16.1	19.3	23.9	26.6	26.4	24.3	19.2	16.0	10.2	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	15.7	17.2	22.0	24.4	27.2	28.7	28.3	27.7	24.5	21.2	16.4	(27)	(28)
		MCDB	16.1	18.8	21.1	26.1	28.8	32.4	33.8	33.2	31.6	27.3	23.4	17.4	(28)	(29)
	2%	WB	11.1	12.8	15.6	20.3	23.0	26.4	28.0	27.7	26.6	22.7	19.3	13.5	(29)	(30)
		MCDB	12.5	15.3	19.2	24.0	26.2	31.0	32.9	32.2	29.9	24.9	21.1	15.3	(30)	(31)
	5%	WB	9.3	10.8	14.2	18.9	21.9	25.7	27.5	27.2	25.8	21.4	17.8	12.1	(31)	(32)
		MCDB	10.8	12.7	17.5	22.2	24.8	29.6	32.3	31.4	28.9	23.5	19.7	13.8	(32)	(33)
	10%	WB	7.7	9.2	12.7	17.4	20.9	24.8	27.1	26.9	25.1	20.4	16.5	10.5	(33)	(34)
		MCDB	9.4	11.1	15.8	20.7	23.7	27.9	31.7	30.8	28.0	22.9	18.4	12.3	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.6	6.7	7.2	6.8	5.3	5.8	5.1	4.9	5.4	6.0	5.9	(35)	(36)
		MCDBR	8.6	10.0	11.1	11.0	10.1	8.3	7.7	7.1	6.2	6.7	7.7	8.1	(36)	(37)
	5% WB	MCWBR	5.6	6.1	5.7	5.5	4.2	3.1	2.4	2.3	2.4	3.0	4.1	5.2	(37)	(38)
		MCWBR	6.5	8.1	9.2	8.7	6.9	6.8	6.7	5.9	5.4	5.0	5.8	5.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.590	0.631	0.664	0.671	0.657	0.715	0.689	0.654	0.602	0.603	0.614	0.584	(39)	(40)
		taud	1.677	1.624	1.587	1.629	1.694	1.627	1.713	1.768	1.844	1.758	1.688	1.702	(40)	(41)
	Ebn at Noon		599	626	648	670	685	644	659	675	689	642	575	575	(41)	(42)
		Edh at Noon	203	234	259	258	243	259	237	221	198	201	199	189	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2.27	2.65	3.78	4.44	4.78	4.18	5.25	4.88	3.99	3.39	2.51	2.22	(43)	(44)
	RadStd		0.29	0.51	0.32	0.51	0.52	0.61	0.63	0.66	0.39	0.45	0.30	0.30	(44)	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	N/A

Nomenclature: See separate page